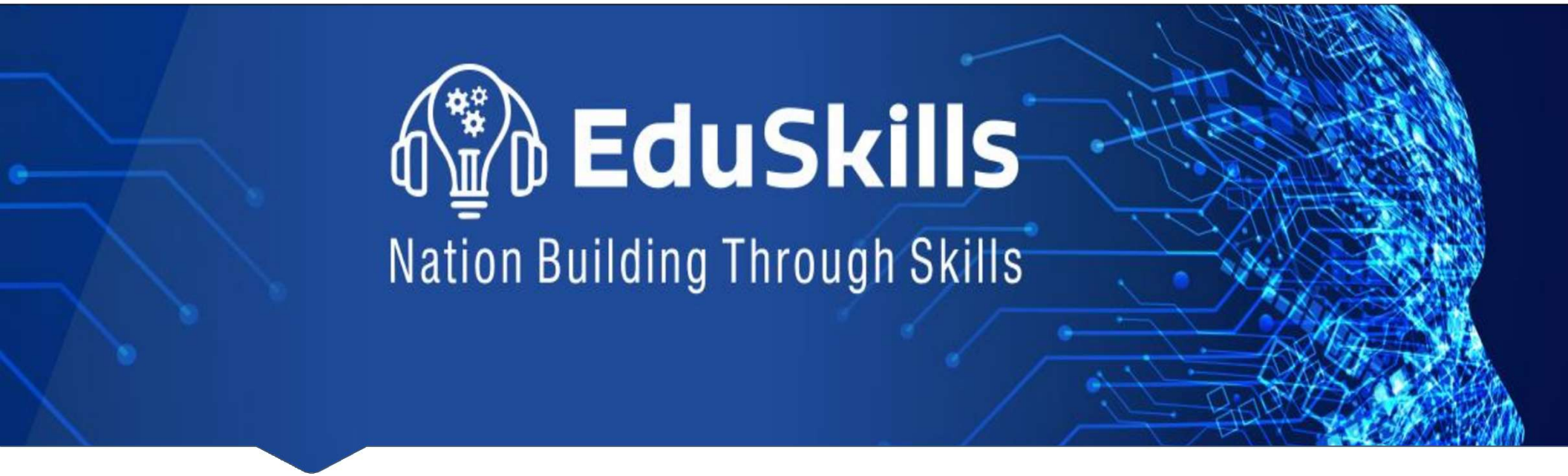
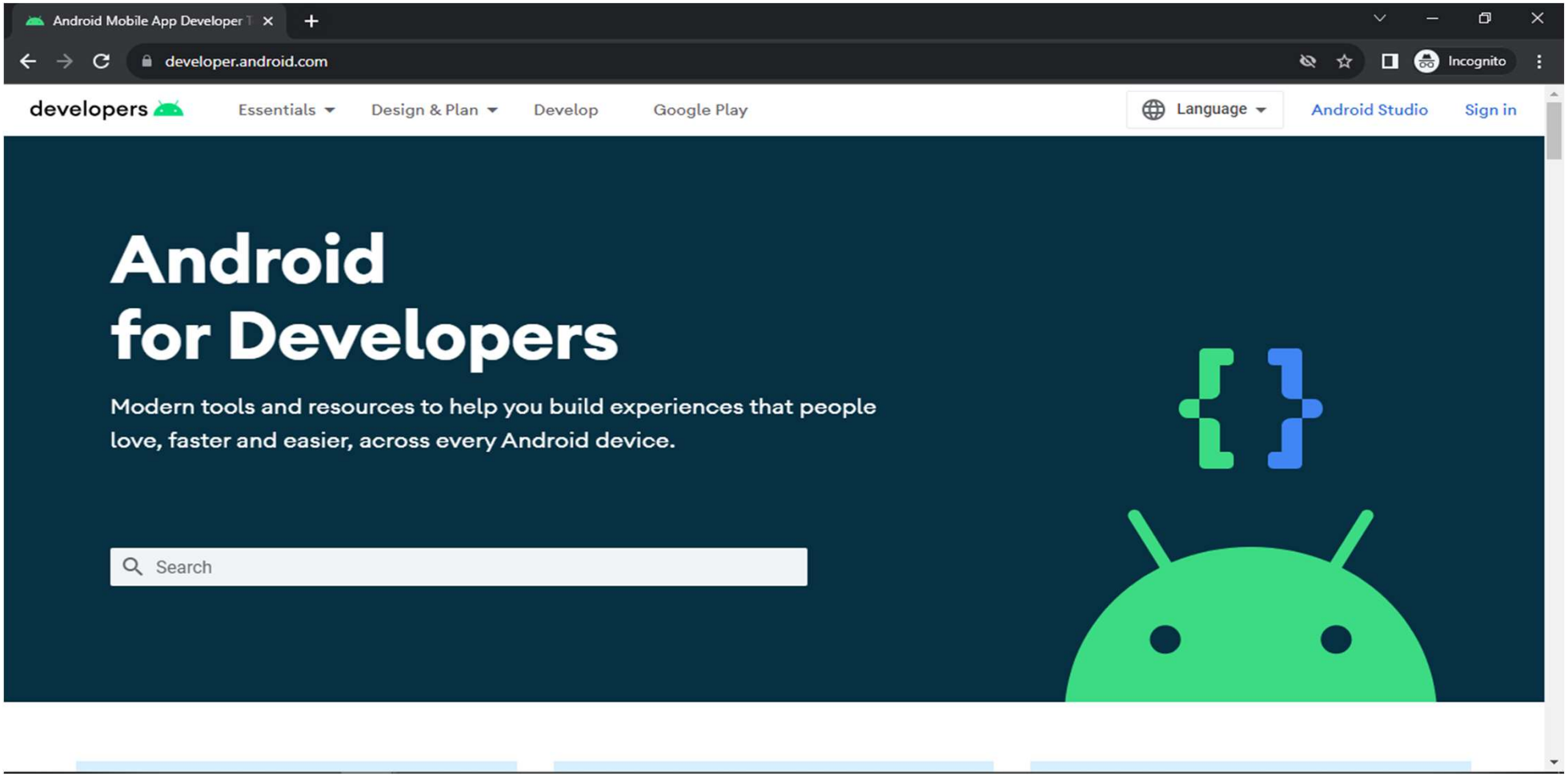


Google for Developers

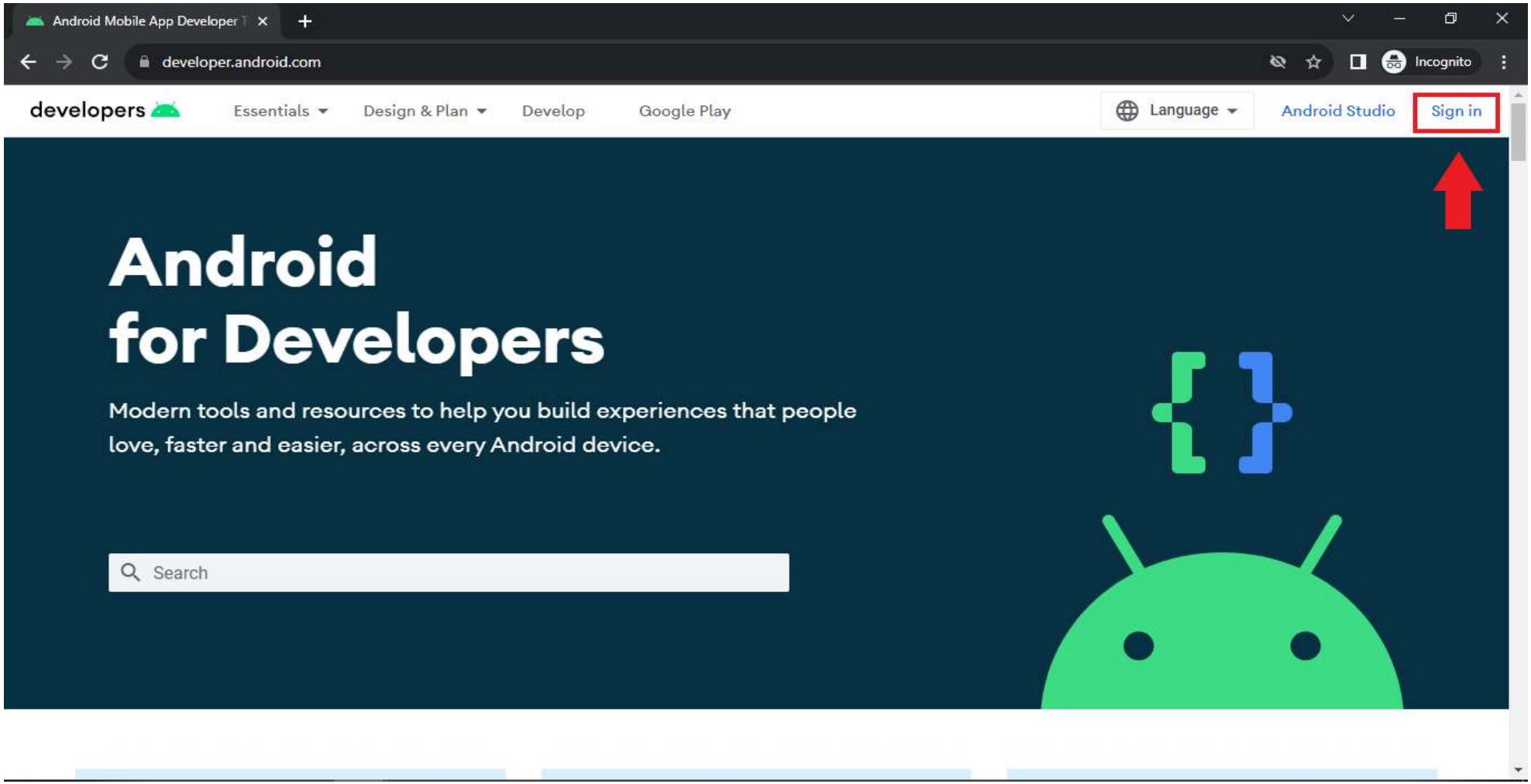
Android Basics with Compose course process document

Part-1 Registration

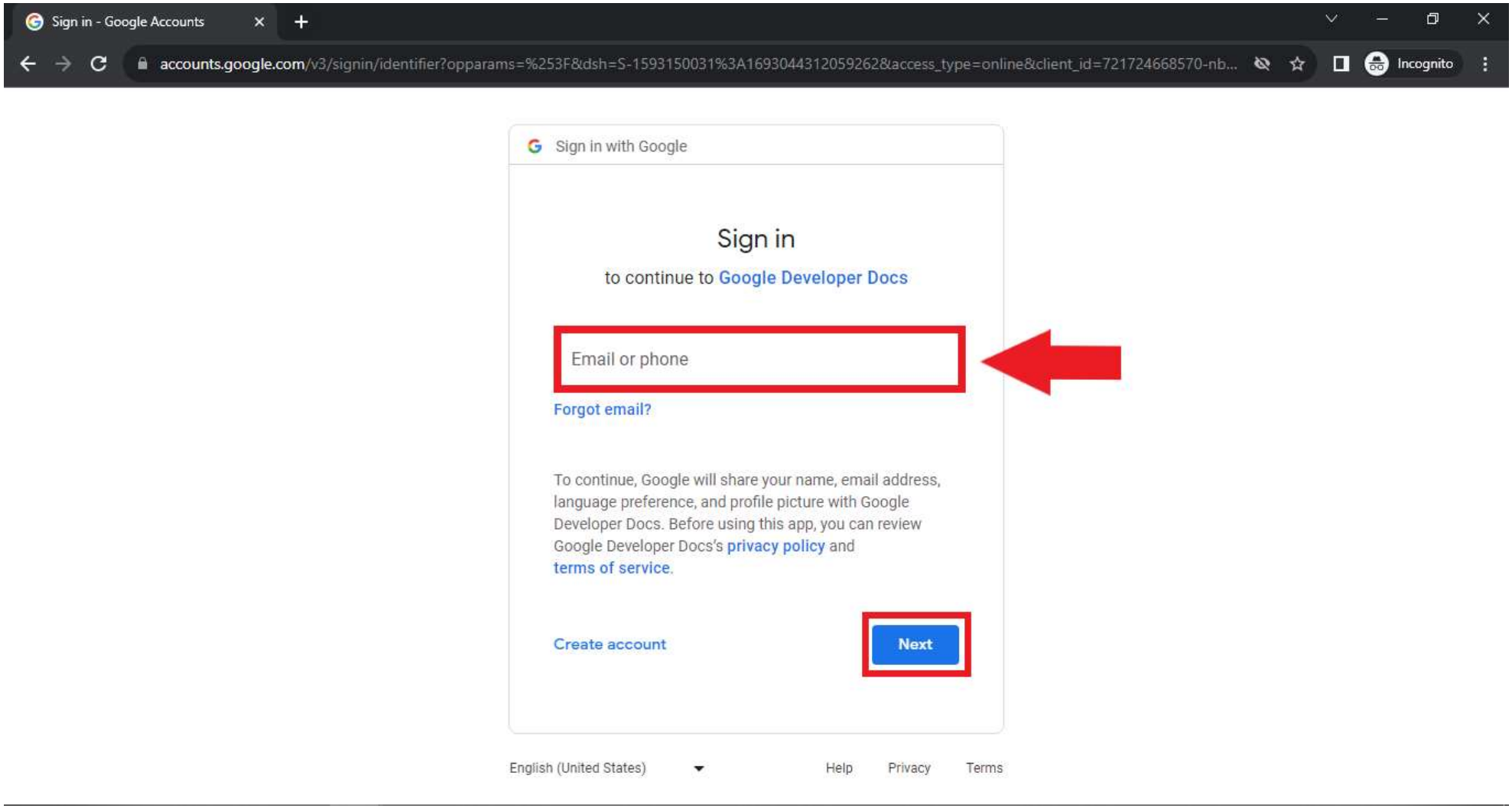
Step: 1 Click on the link given below:
https://developer.android.com/



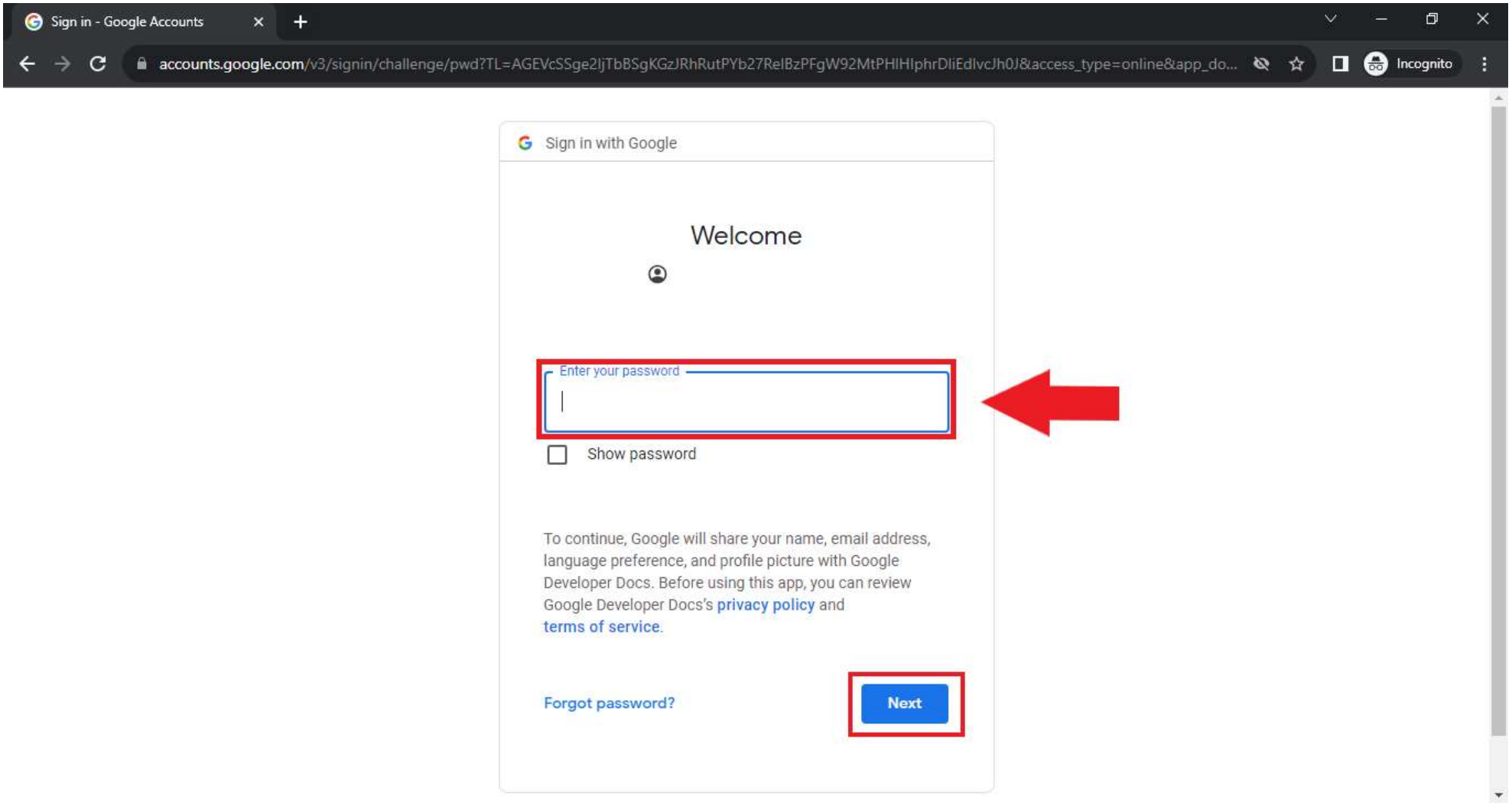
Step: 2 Click on Sign In



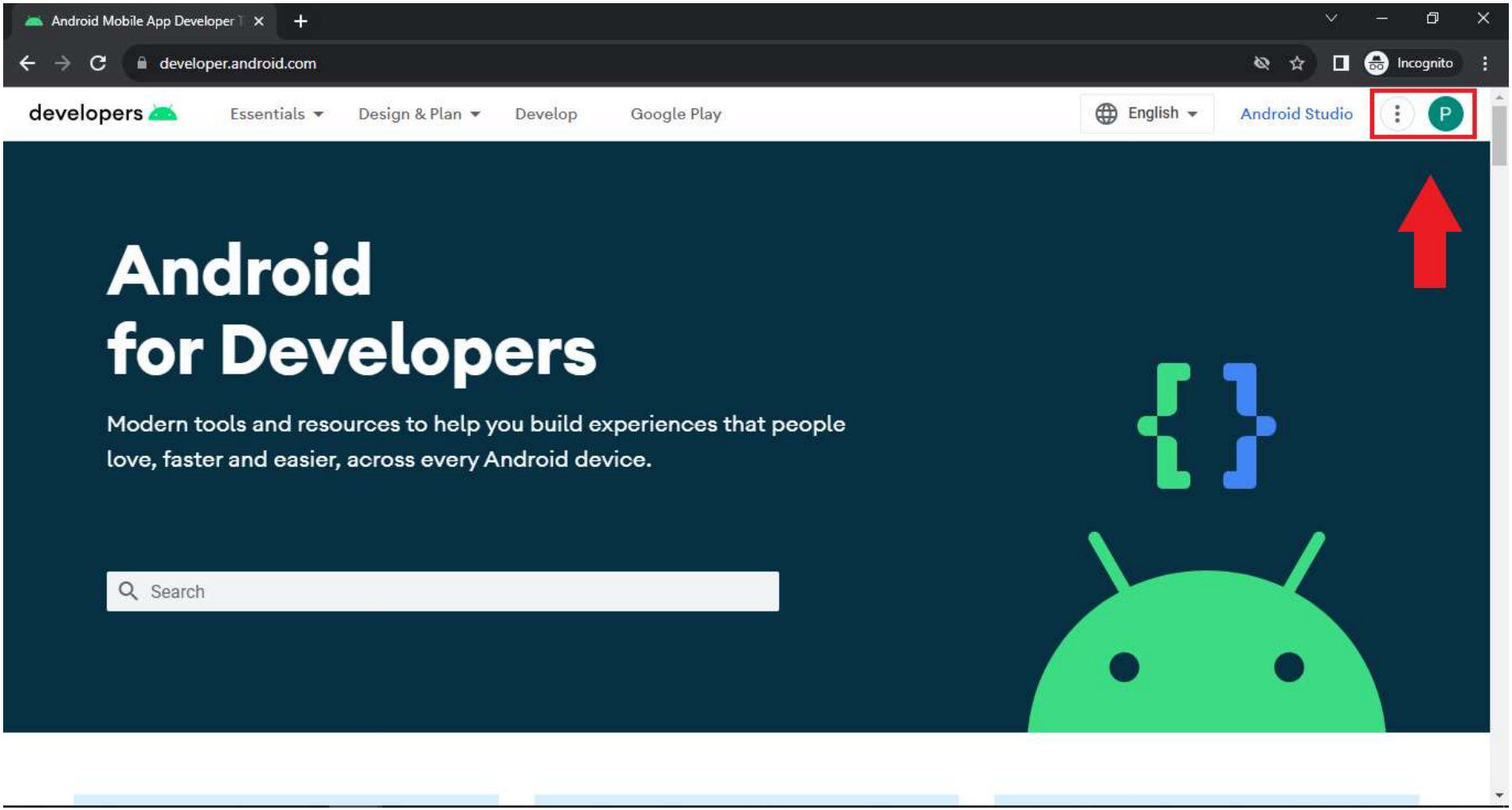
Step: 3 Enter Gmail ID and Click on Next.



Step: 4 Enter Password and Click on Next.

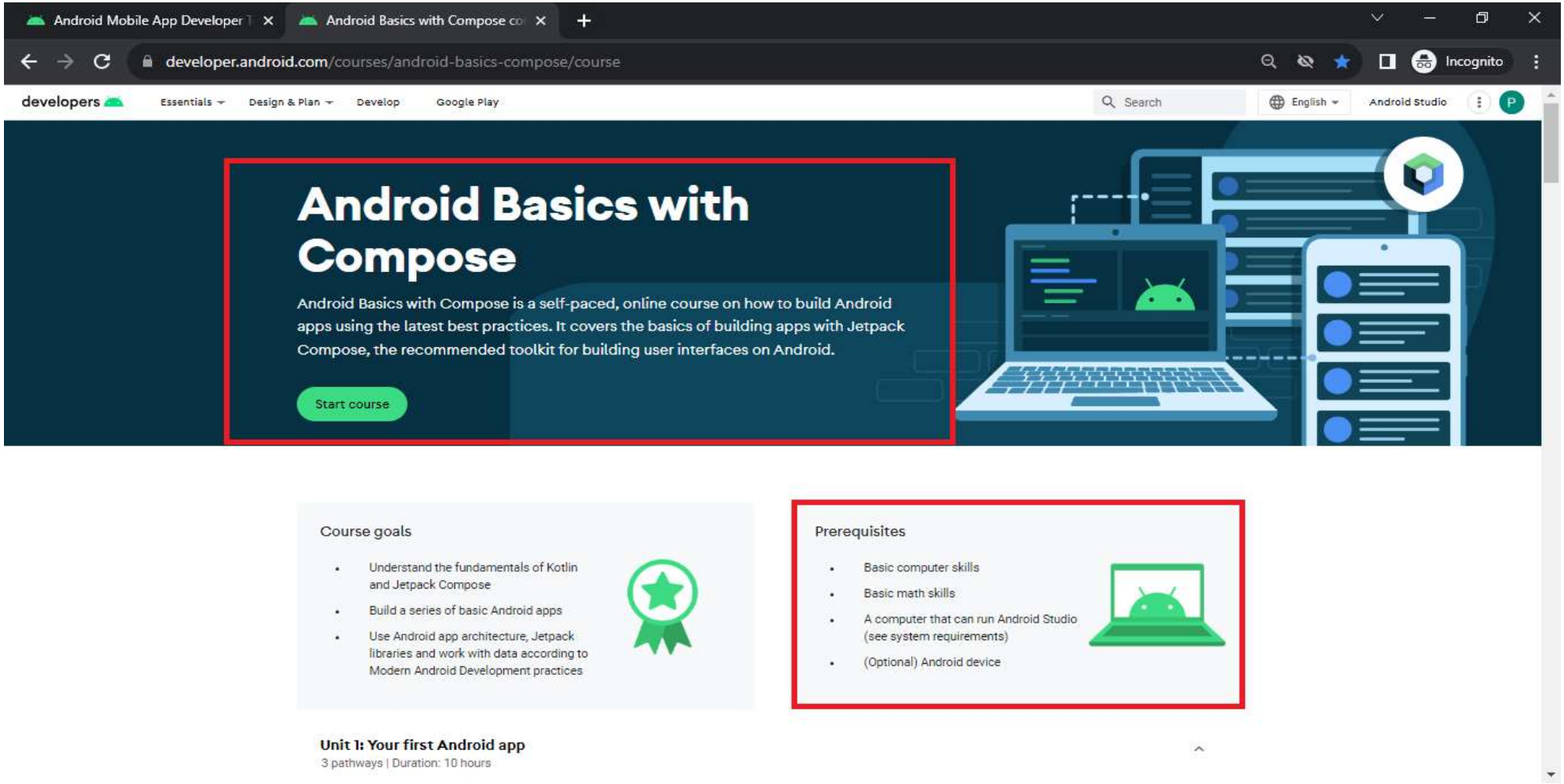


Registration Completed

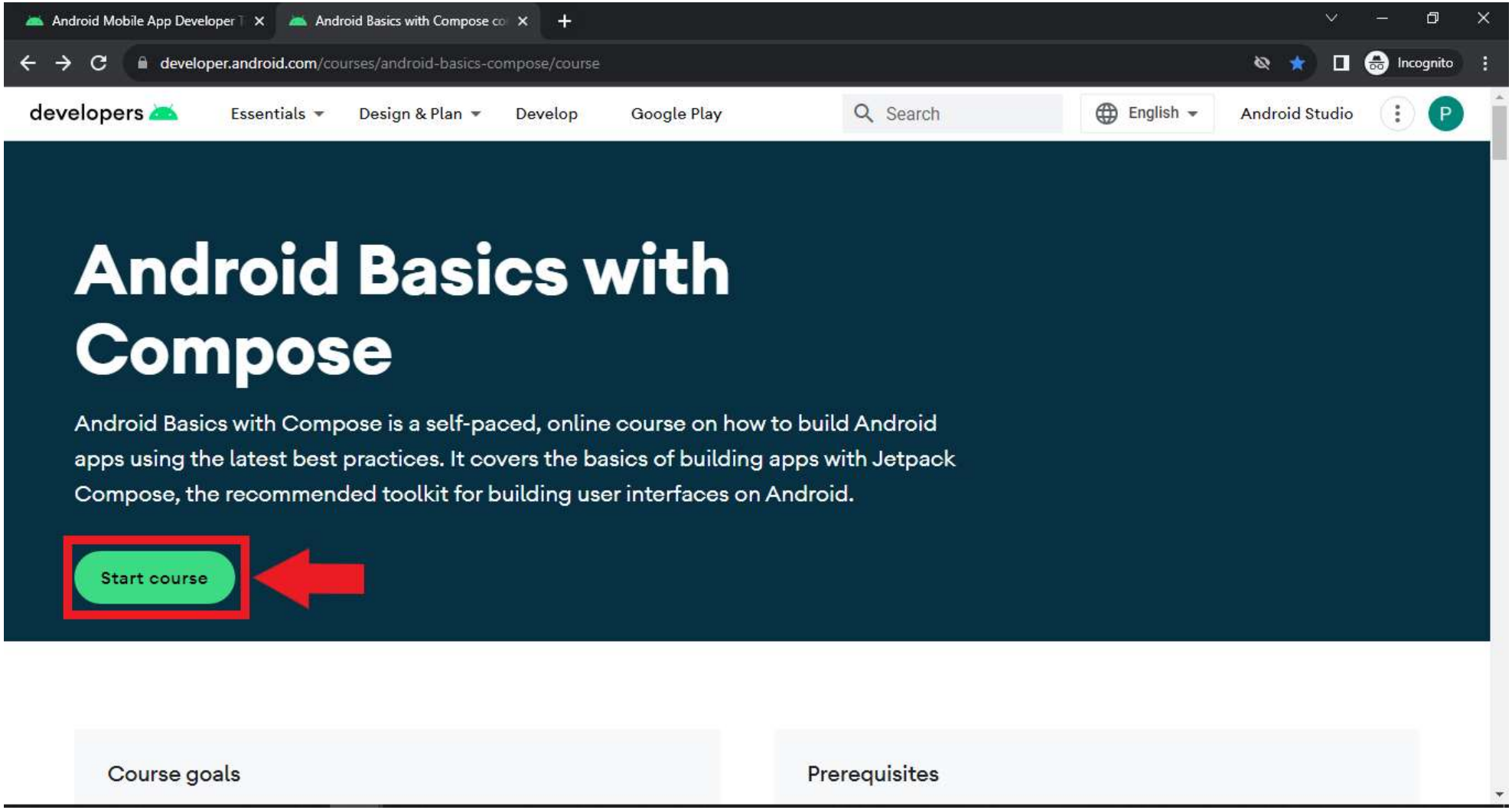


Part-2 Course Enrollment

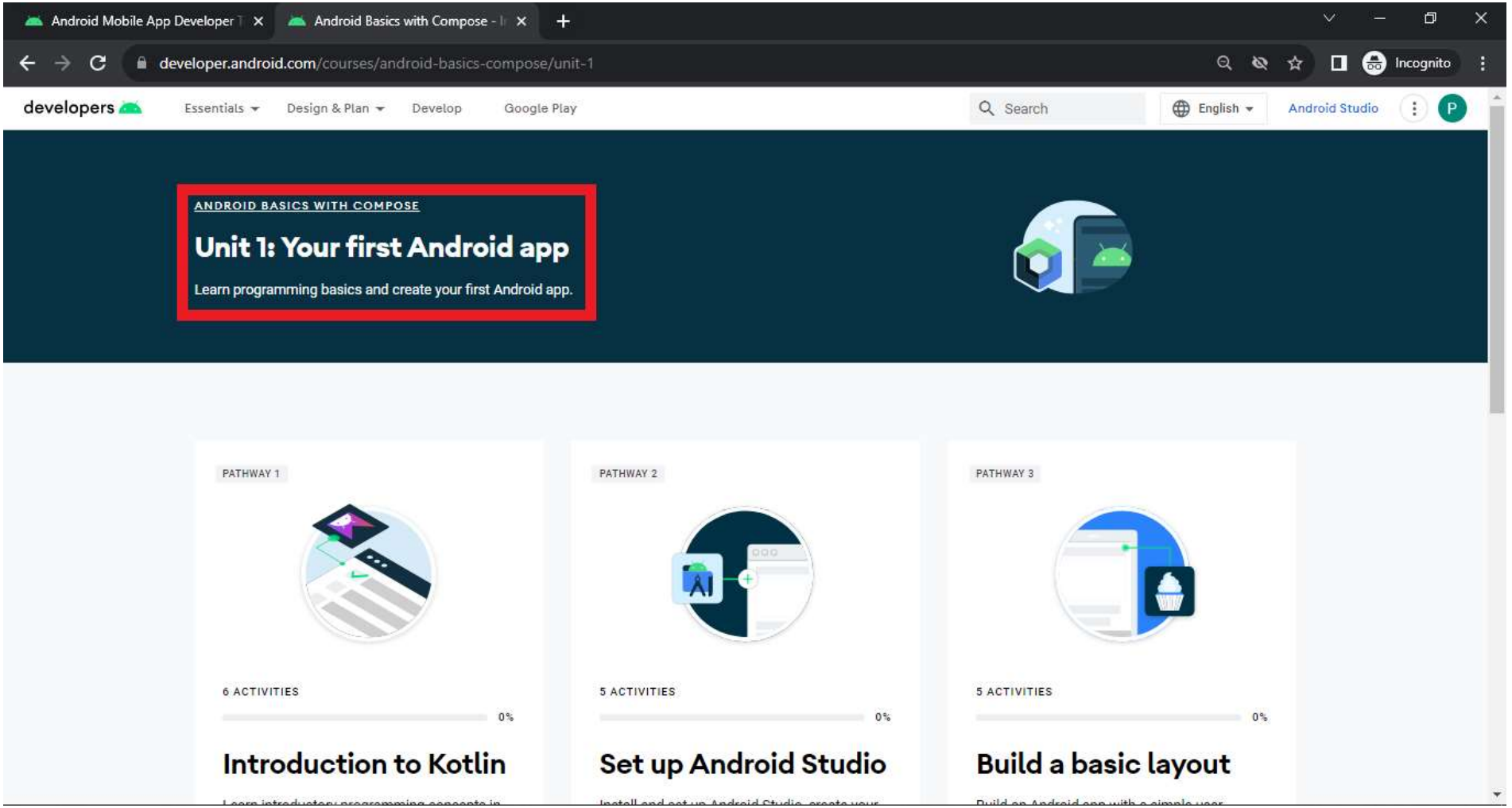
Step: 5 Click on the course link given below:
https://developer.android.com/courses/android-basics-compose/course



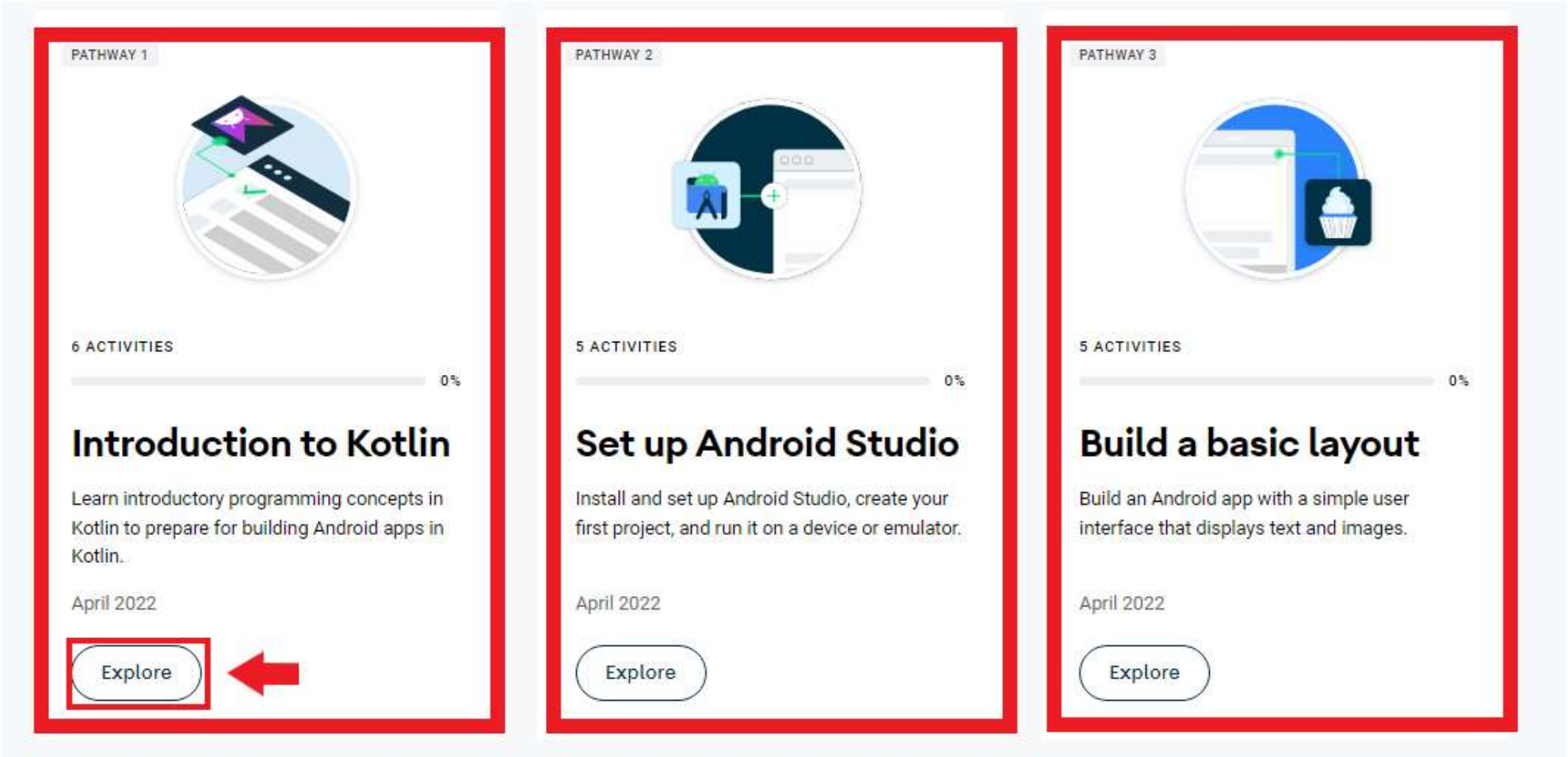
Step: 6 Click on Start course



Step: 7 Observe the pathways provided, scroll down to explorer the pathways



Step: 8 Click on Explorer option under pathway and complete the pathway to earn badges



Complete Unit-1

Unit 1: Your first Android app

3 pathways | Duration: 10 hours

Learn programming basics and create your first Android app.

- Write simple Kotlin programs that display text output.
- Download and install Android Studio.
- Build an Android app with a simple user interface that displays text and images.
- Run the app on a device or emulator.



Introduction to Kotlin

Pathway 1 | Duration: 3 hours

Learn introductory programming concepts in Kotlin.



Setup Android Studio

Pathway 2 | Duration: 4 hours

Install and set up Android Studio, create your first project, and run it on a device or emulator.



Build a basic layout

Pathway 3 | Duration: 3 hours

Build an Android app with a simple user interface that displays text and images.

Complete Unit-2

Unit 2: Building app UI

3 pathways | Duration: 21 hours

Continue learning the fundamentals of Kotlin, and start building more interactive apps.

- Use conditionals, function types, classes, and lambda expressions in Kotlin.
- Understand how composition and recomposition works.
- Add a button to an app UI and respond to user taps.
- Create an app that works with data entered by the user.
- Learn how to use state to display data and reflect the changes automatically when the data gets updated.
- Write unit tests to test isolated functions.



Kotlin fundamentals

Pathway 1 | Duration: 8 hours

Learn more fundamentals of Kotlin, object-oriented programming, and lambdas.



Add a button to an app

Pathway 2 | Duration: 6 hours

Learn how to respond to a button click in an Android app.



Interacting with UI and state

Pathway 3 | Duration: 7 hours

Create a tip calculator app that calculates the tip from user input.

Complete Unit-3

Unit 3: Display lists and use Material Design

3 Pathways | Duration: 15 hours

Build apps that display a list of data and learn how to make your apps more beautiful with Material Design.

- Use data classes, functions, and collections in Kotlin.
- Create a scrollable list in an app that displays both text and images.
- Add click listeners to interact with list items.
- Add an app bar to the app and modify the app theme.
- Use Material Design to build modern and intuitive user interfaces, using colors, shapes and typography.



More Kotlin fundamentals

Pathway 1 | Duration: 8 hours

Learn additional Kotlin programming concepts that will enable you to build more interesting and fun Android apps.



Build a scrollable list

Pathway 2 | Duration: 4 hours

Create an app that displays a scrollable list of text and images using Compose.



Build beautiful apps

Pathway 3 | Duration: 3 hours

Make your apps more beautiful and intuitive to use with Material Design, animations, and accessibility best practices.

Complete Unit-4

Unit 4: Navigation and app architecture

3 pathways | Duration: 27 hours

Learn the best practices of app architecture to build more complex apps. Enhance your users' ability to navigate across, into and back out from the various screens within your app for a consistent and predictable user experience.

- Explain activities and their lifecycles.
- Understand Modern Android architecture.
- Use StateFlow and UDF pattern to work with state and events.
- Add a ViewModel to save data and state.
- Set up and use the Navigation component with Compose.
- Understand what responsive UI is.
- Use window class sizes to build for different screen sizes.
- Add a navigation drawer to an app.



Architecture Components

Pathway 1 | Duration: 9 hours

Learn about app architecture and how to use ViewModels, UI State, and StateFlow to build more complex apps.



Navigation in Jetpack Compose

Pathway 2 | Duration: 6 hours

Learn how to use the Navigation component to build more complex apps with more screens and how to navigate and pass data between different composables.



Adapt for different screen sizes

Pathway 3 | Duration: 13 hours

In this pathway, you'll learn how to adapt your app to different screen sizes and provide a better user experience, as well as how to test your adaptive UI.

Complete Unit-5

Unit 5: Connect to the internet

2 Pathways | Duration: 14 hours

Use Kotlin coroutines to perform multiple tasks at once, and learn about HTTP and REST to get data from the internet using Retrofit. Then use the Coil library to display images in your app.

- Describe the basics of concurrency and how to use coroutines in an Android app.
- Define and understand the data layer in Modern Android app architecture.
- Implement a repository to centralize data access.
- Use Retrofit to retrieve data from a remote server.
- Load and display images using the Coil library.
- Implement dependency injection to decouple the classes, making it easier to test, maintain, and scale the app.



Get data from the internet

Pathway 1 | Duration: 6 hours

Implement coroutines to perform tasks concurrently without blocking the app, and learn about HTTP and REST to get data from the internet.



Load and display images from the internet

Pathway 2 | Duration: 3 hours

Apply architecture best practices to the app and use Coil to download and display images.

Complete Unit-6

Unit 6: Data persistence

3 Pathways | Duration: 10 hours

Learn how to store data locally on the device and keep your apps working through any network disruptions for a smooth and consistent user experience.

- Learn the basics of SQL to insert, update, and delete data from a SQLite database.
- Use the Room library to add a database to an Android app.
- Use Database Inspector to test and debug database issues.
- Use Preference DataStore to store user preferences.



Introduction to SQL

Pathway 1 | Duration: 2 hours

Learn how to use SQL to read and manipulate data in a relational database.



Use Room for data persistence

Pathway 2 | Duration: 6 hours

Use the Room library to easily create and use relational databases in an Android app.



Store and access data using keys with DataStore

Pathway 3 | Duration: 2 hours

Learn how to store simple, key-value pair data with Preferences DataStore in an Android app

Complete Unit-7

Unit 7: WorkManager

1 Pathway | Duration: 6 hours

Use Android Jetpack's WorkManager API to schedule necessary background work, such as data backups or fresh content downloads, that keeps running even if the app exits or the device restarts.

- Define long running tasks that need to run in background work.
- Add WorkManager to an Android app.
- Create a Worker object and enqueue work.
- Create constraints on WorkRequests.
- Use the Background Task Inspector to inspect and debug WorkManager.



Schedule tasks with WorkManager

Pathway 1 | Duration: 3 hours

Learn when and how to use WorkManager, an API that handles background work that needs to run regardless of whether the application process is still running.

Complete Unit-8

Unit 8: Views and Compose

2 Pathways | Duration: 4 hours

Learn how to use Compose and the older UI toolkit based on Views side-by-side in the same app. In this unit, you will learn interoperability APIs and best practices to add a new feature to an existing app in Views, use an existing library that uses Views, or use a UI component that is not yet available in Compose.

- Understand the View-based UI toolkit and build app UI using XML.
- Add a composable in an app built with Views.
- Add Navigation component to the app and use it to navigate between fragments.
- Use AndroidView to display views.
- Add existing View-based UI components in a Compose app.



Android Views and Compose in Views

Pathway 1 | Duration: 3 hours

Learn the basics of building apps with Android Views and how to add a composable in an app built with Views.



Views in Compose

Pathway 2 | Duration: 1 hours

Learn how to add and use existing Views inside an app built with Jetpack Compose.

NOTE

1.There are 8 units in Android Basics with Compose course.

2.Open 8 units and explorer the pathways to complete the course.

Badges to be earned:

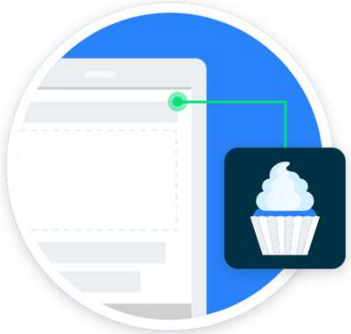
Unit-1



Introduction to Kotlin



Set up Android Studio



Build a basic layout

Note: Sample badge images

Badges to be earned:

Unit-2



Kotlin fundamentals



Add a button to an app



Interact with UI and state

Note: Sample badge images

Badges to be earned:

Unit-3



More Kotlin fundamentals



Build a scrollable list



Build beautiful apps

Note: Sample badge images

Badges to be earned:

Unit-4



Architecture Components



Navigation in Jetpack Compose



Adapt for different screen sizes

Note: Sample badge images

Badges to be earned:

Unit-5



Get data from the internet



Load and display images from the internet

Note: Sample badge images

Badges to be earned:

Unit-6



Introduction to SQL



Use Room for data persistence



Store and access data using keys with DataStore

Note: Sample badge images

Badges to be earned:

Unit-7



Schedule tasks with WorkManager

Note: Sample badge images

Badges to be earned:

Unit-8



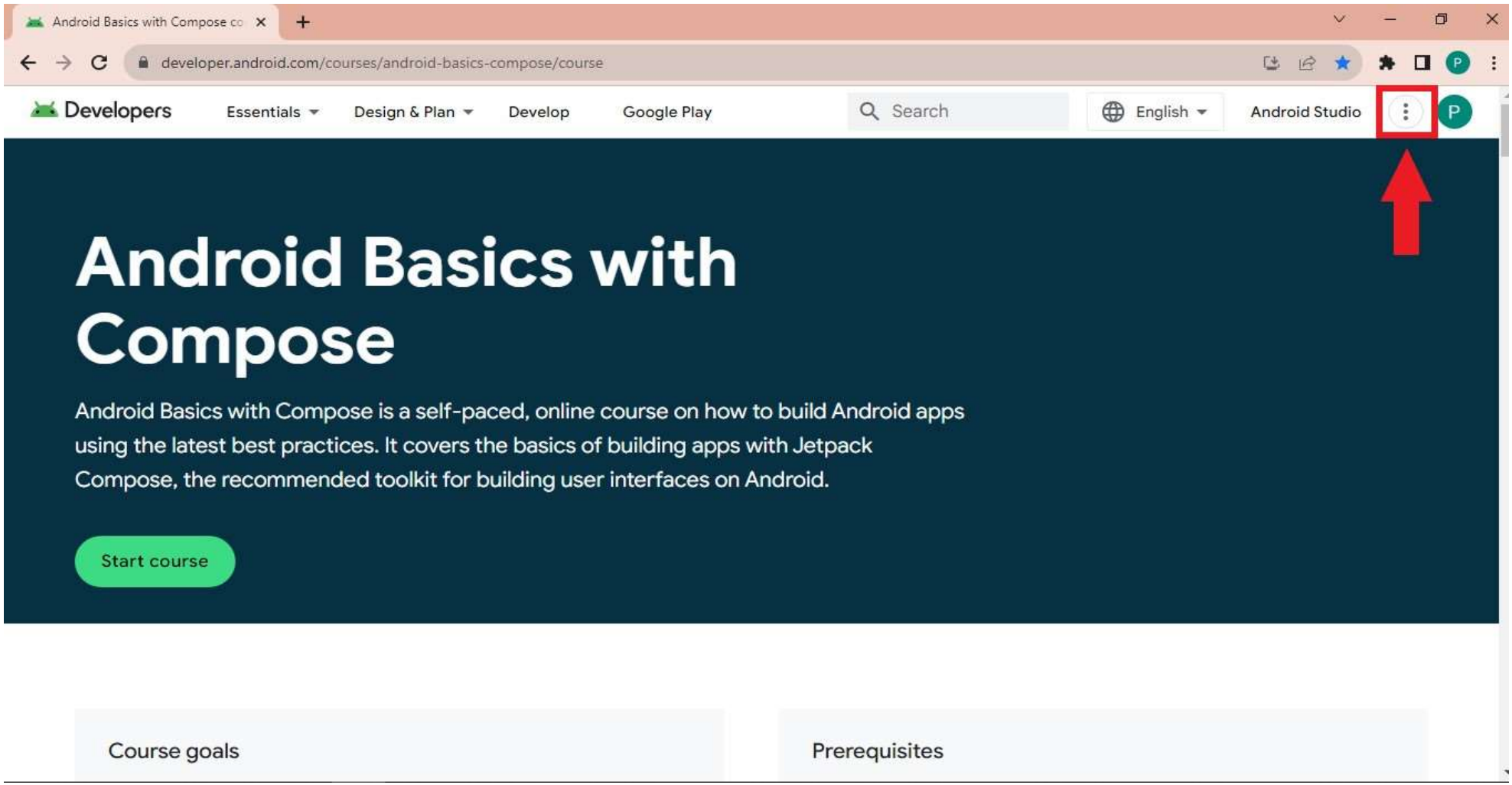
Android Views and Compose in Views



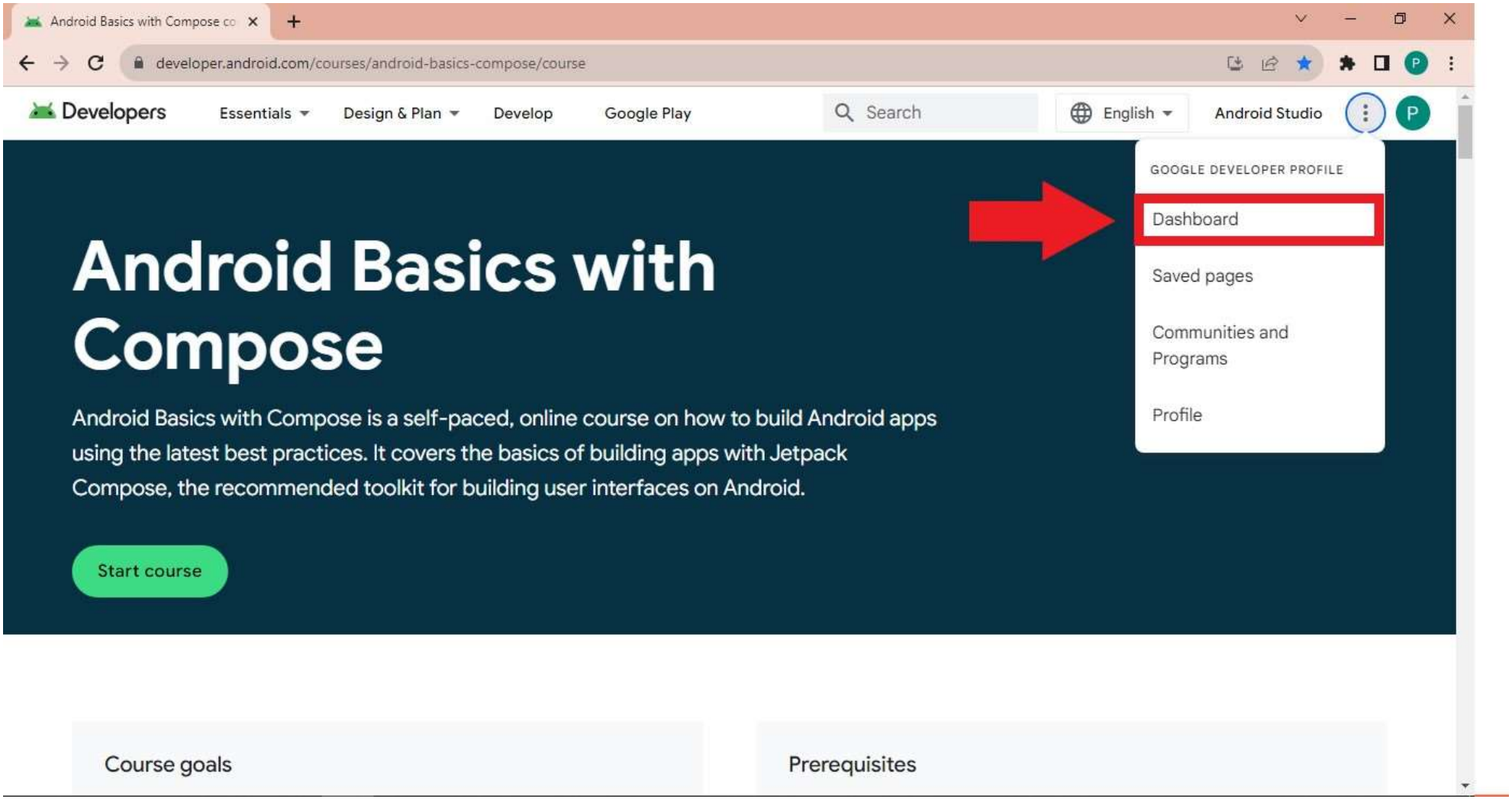
Views in Compose

Note: Sample badge images

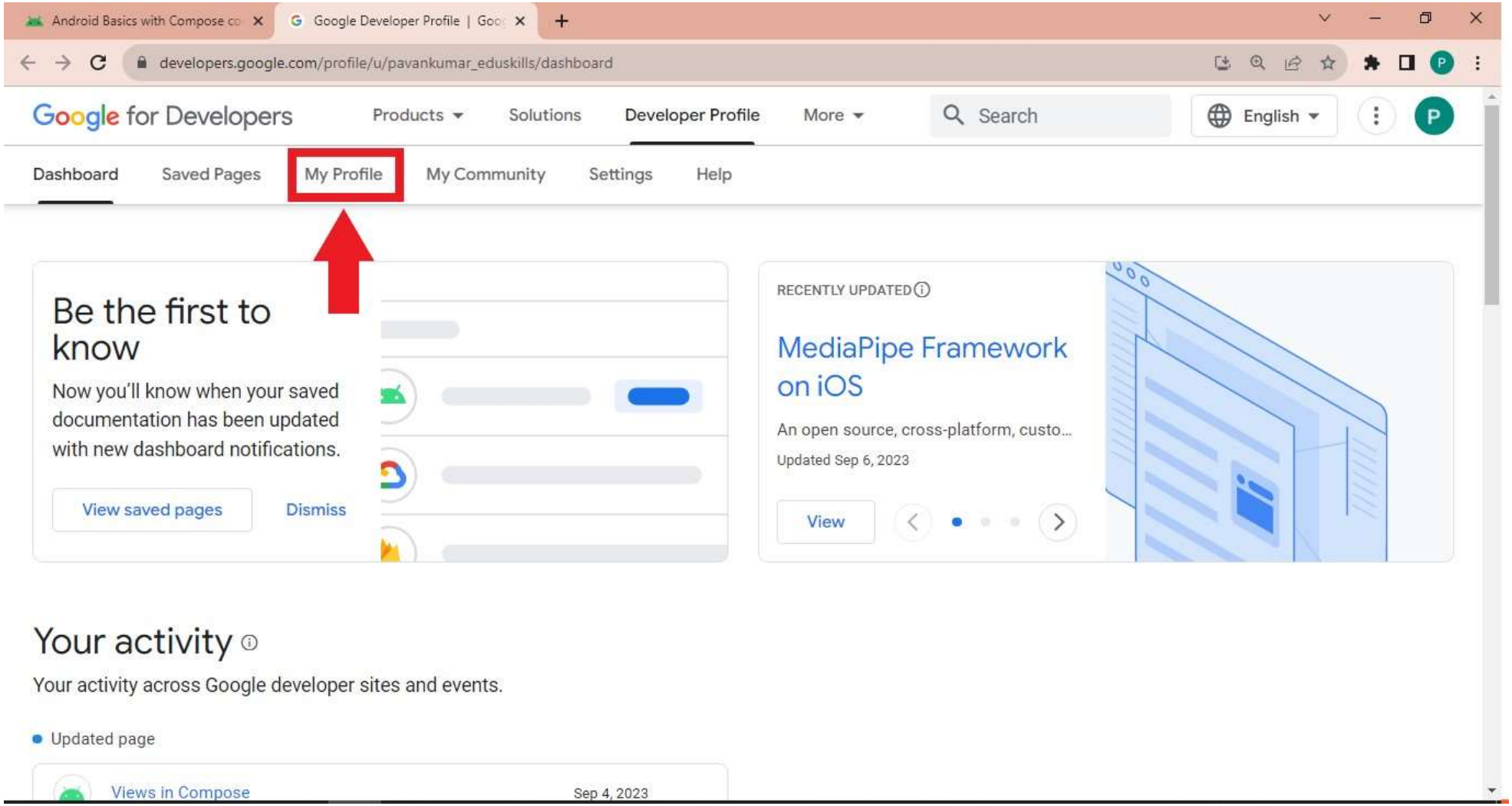
Step: 9 Click on the options as show below.



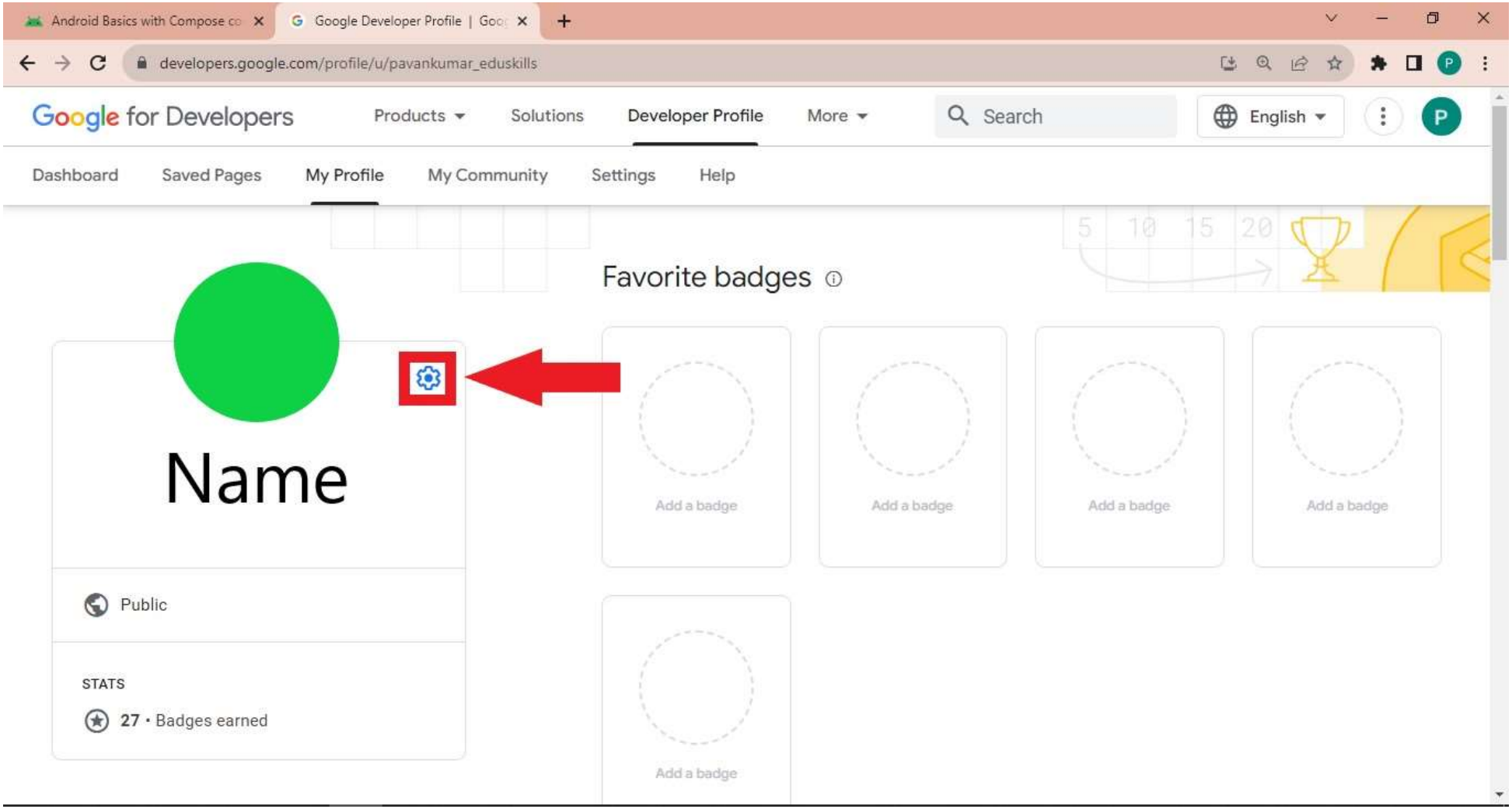
Step: 10 Select dashboard.



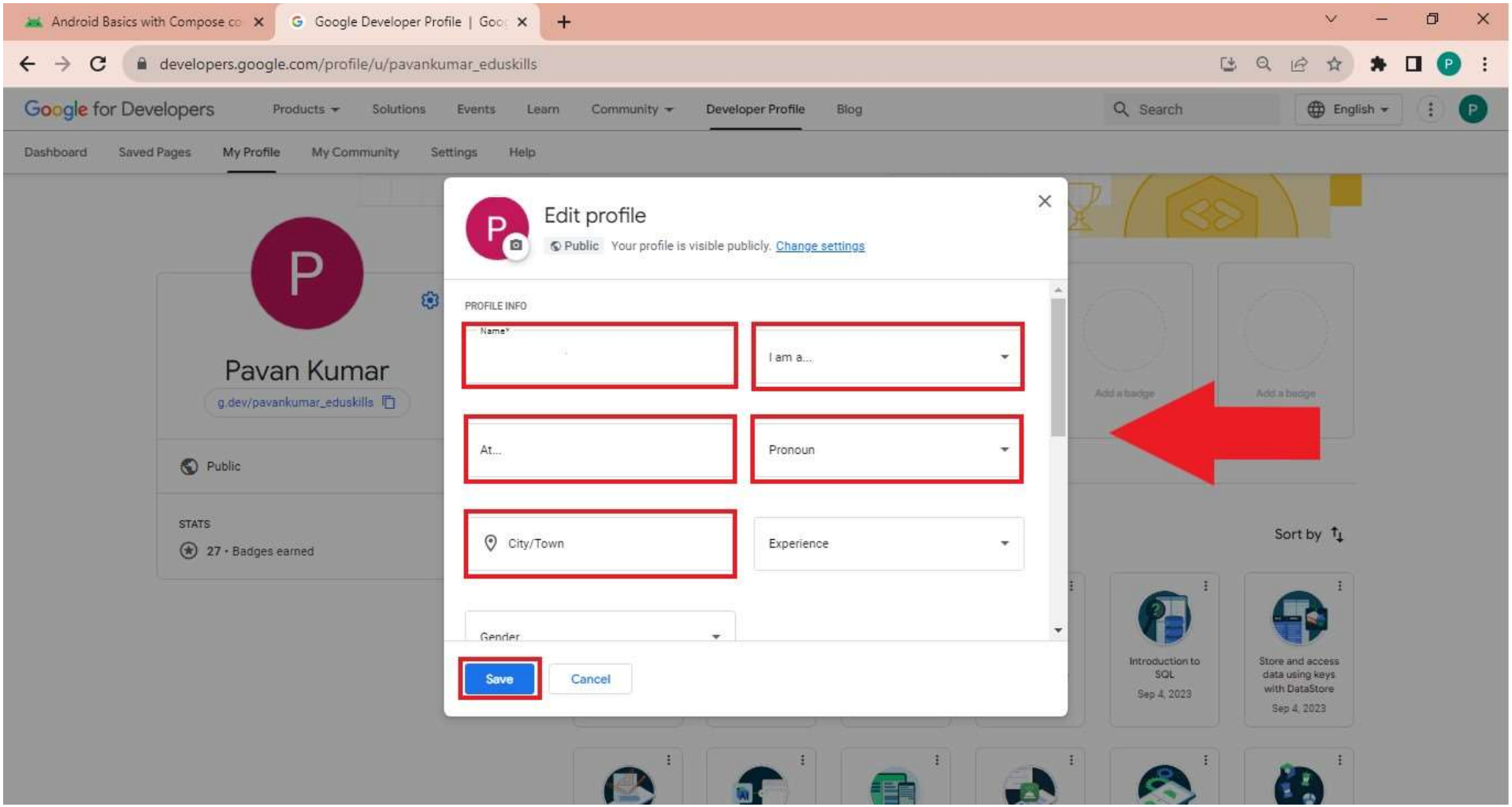
Step: 11 Click on **My Profile** option.



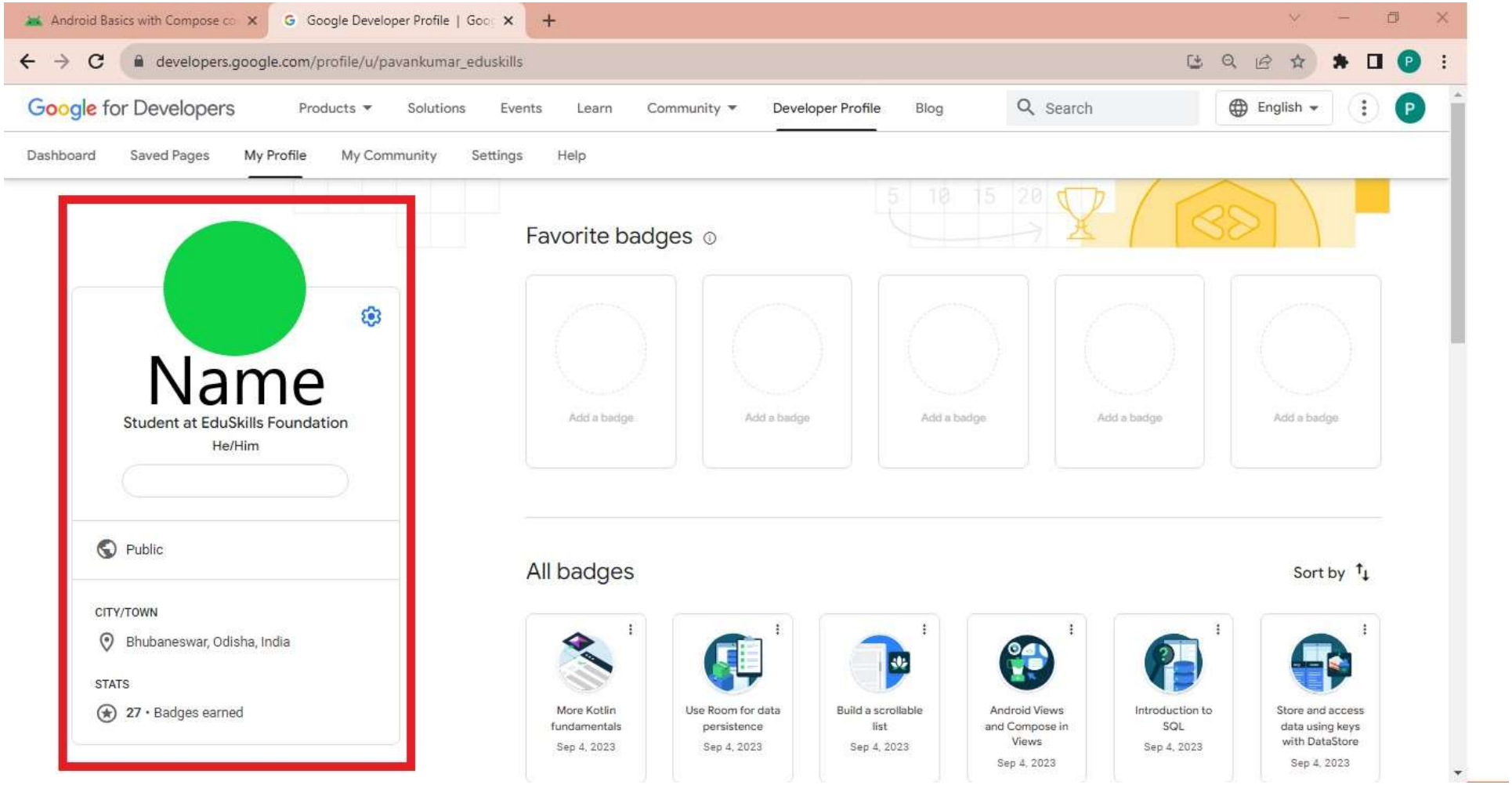
Step: 12 Click on **Settings icon** option.



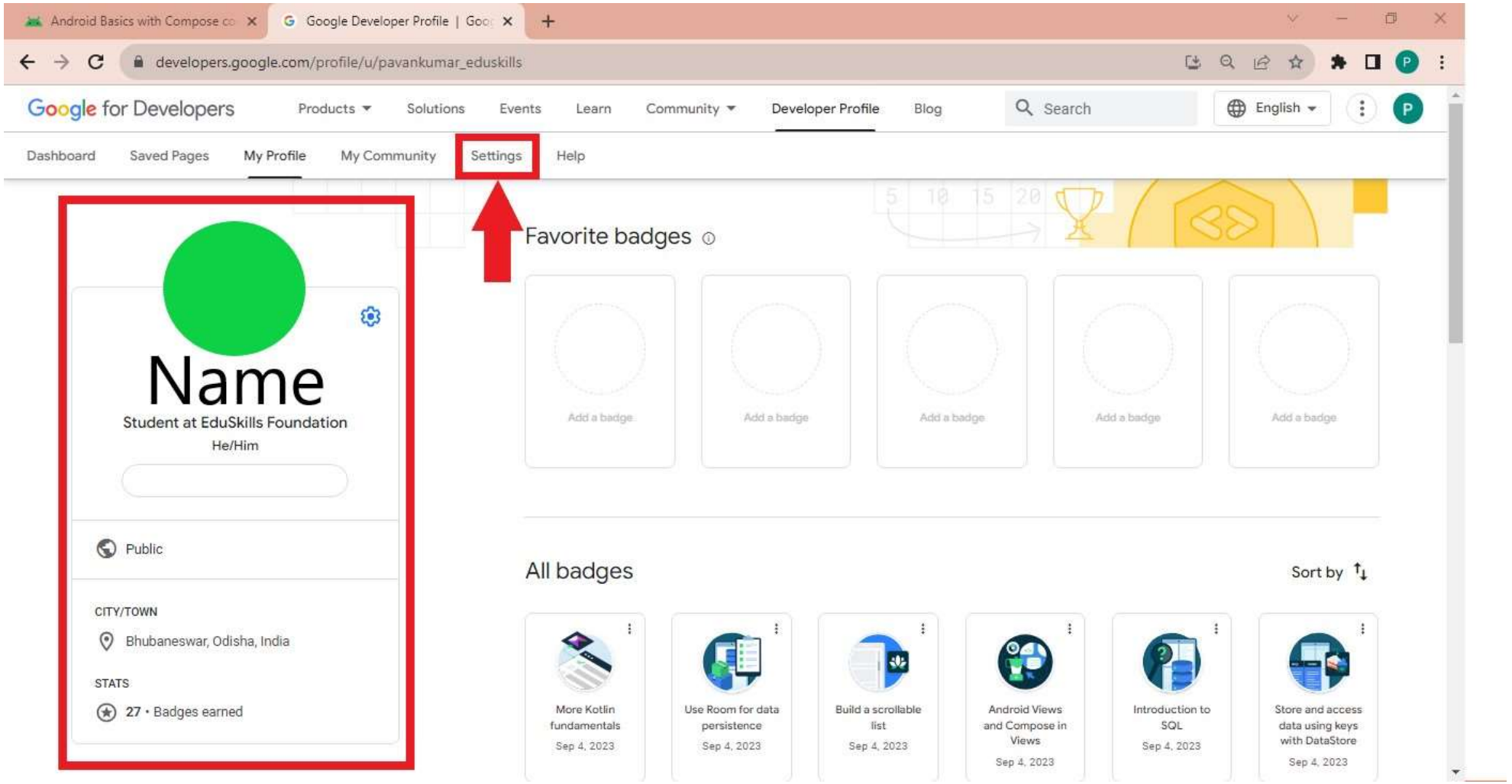
Step: 13 Enter the details



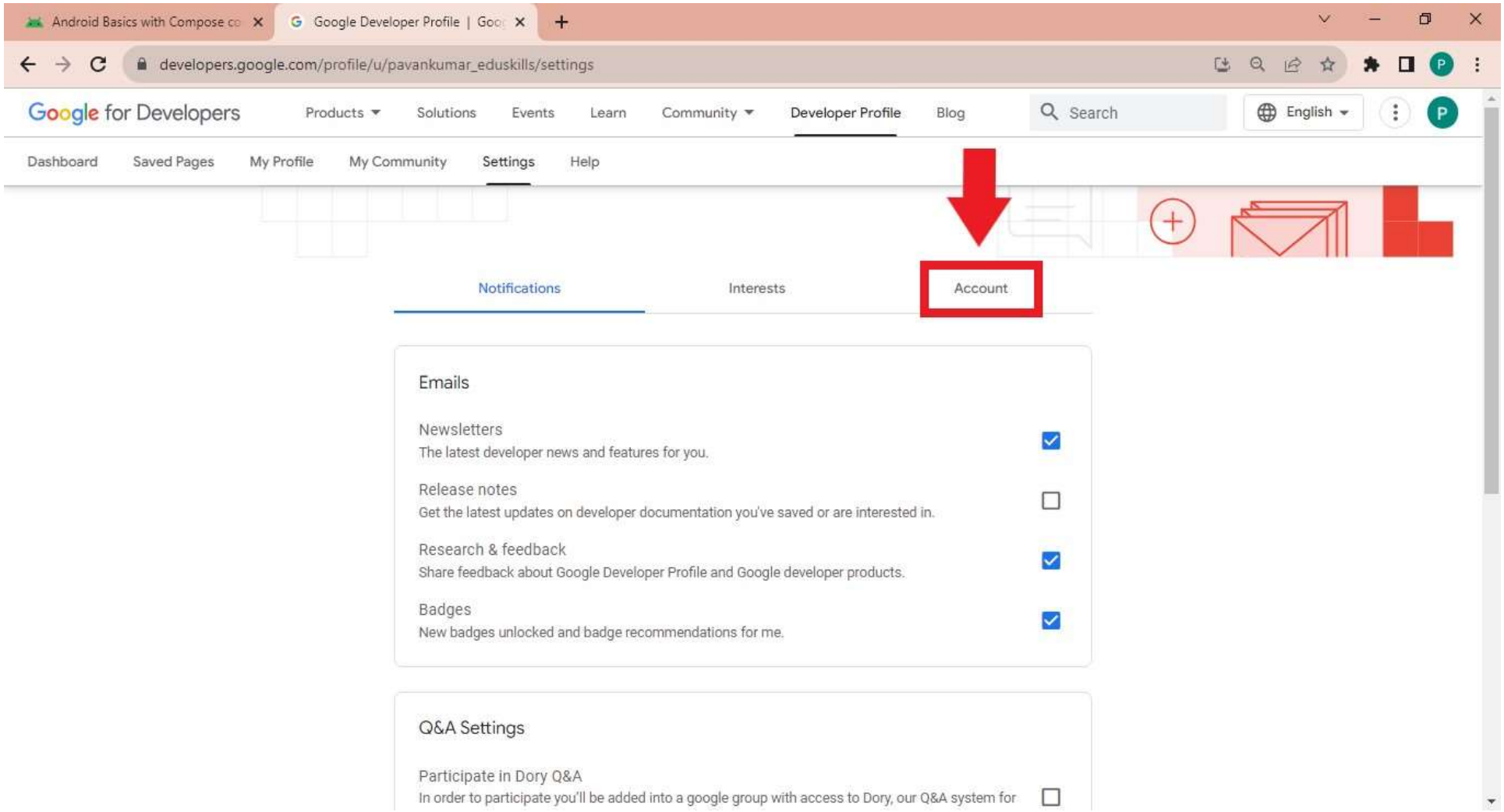
Step: 14 Verify the details. [EXAMPLE]



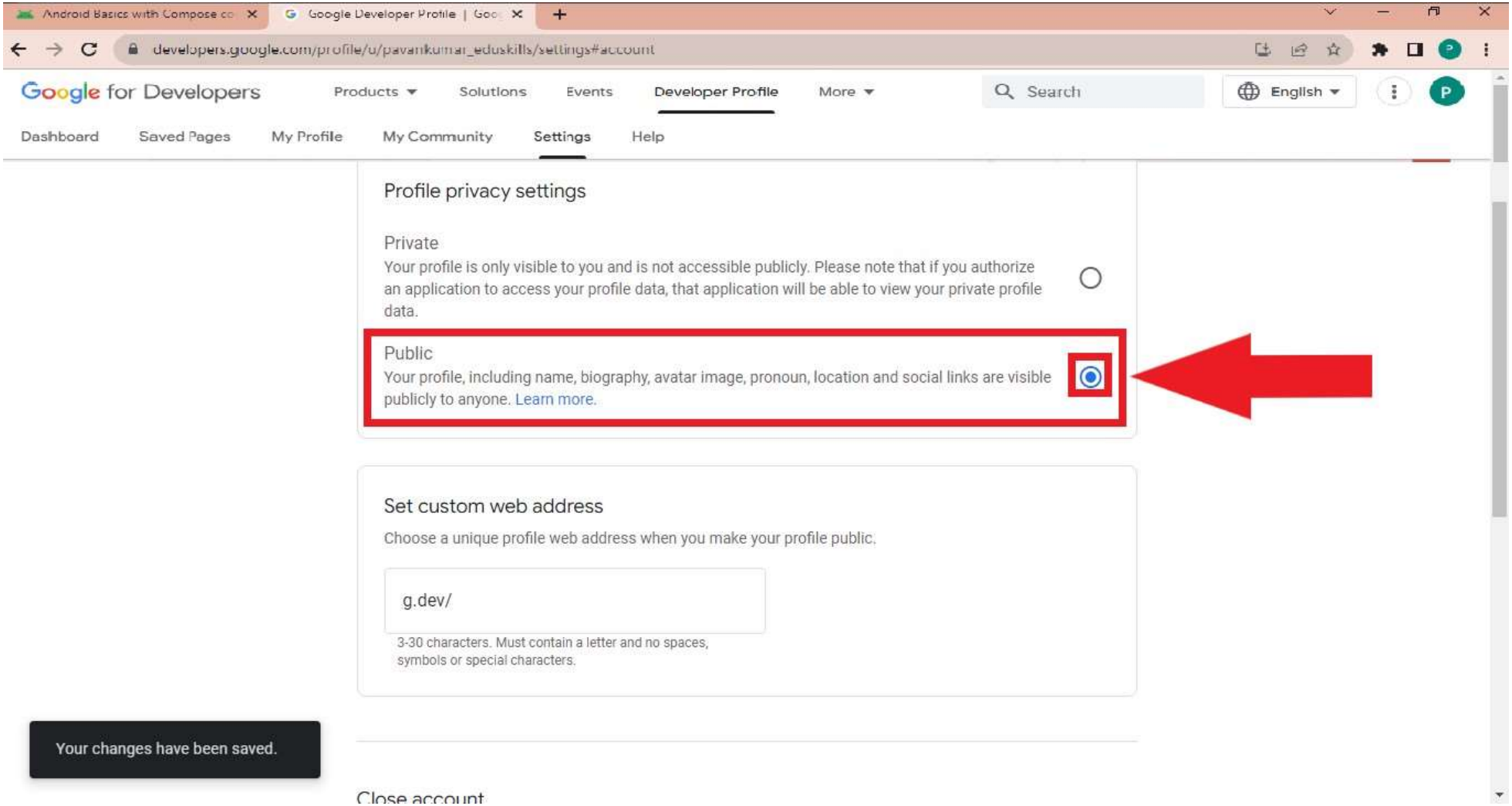
Step: 15 Click on **Settings** option.



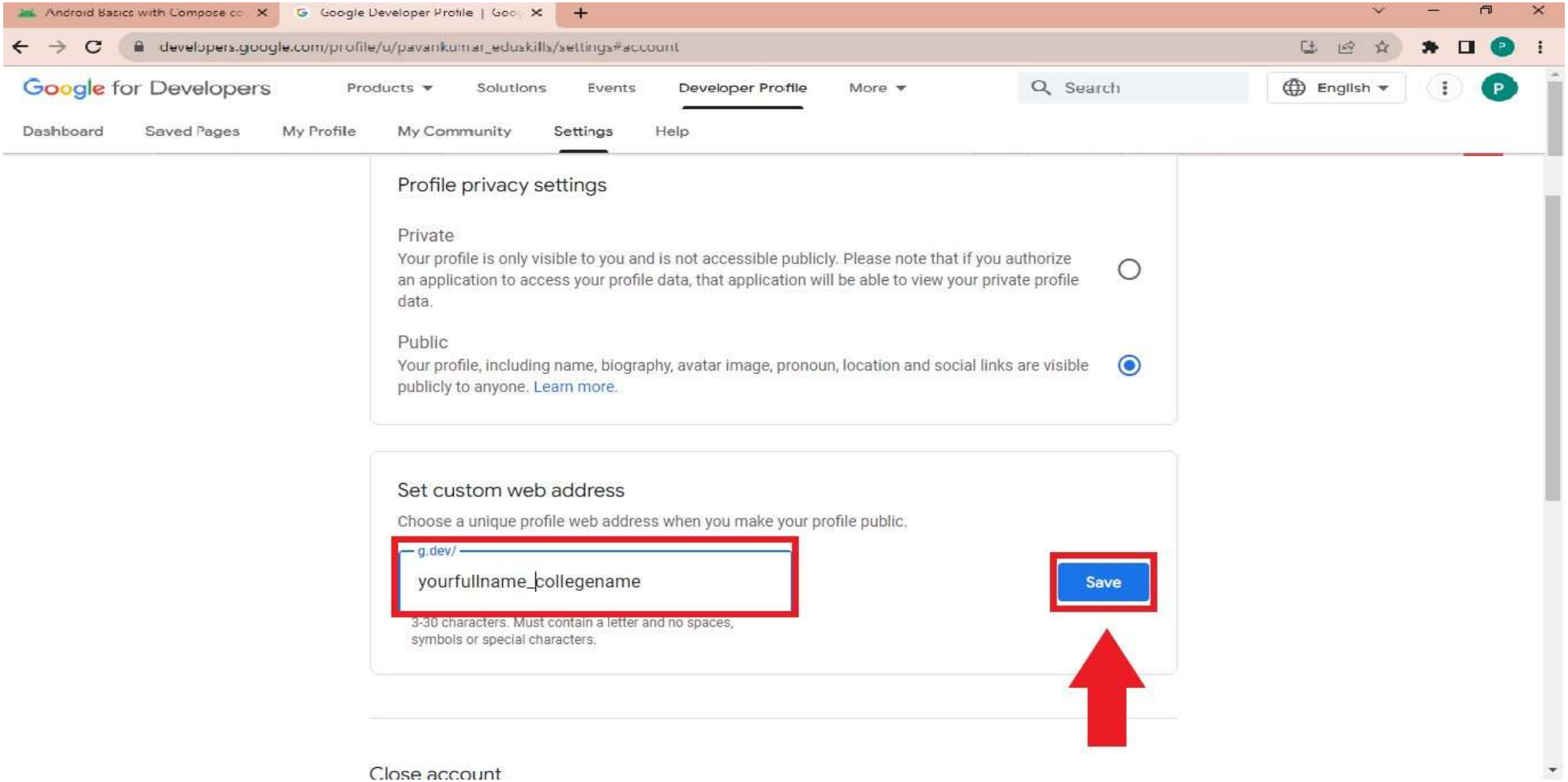
Step: 16 Click on **Account** tab.



Step: 17 Click on **Public radio button**.



Step: 18 Enter **your name & college name** as shown below and click on **save option**.



NOTE

Step:18,Creates a google developer public profile to show case the google badges on online webpage.

Example : ravikumar_ramcollege

URL would be: https://g.dev/ravikumar_ramcollege

Search the URL in google to see the profile and share it for internship evaluation.

Above given example doesn't exist

How to get AICTE internship certficate ?

Ans: Follow the steps given below:

1. Open the given link.

LINK: https://internshipadmin.eduskillsfoundation.org/

2. Enter the AICTE internship registered mail ID and submit the Public Google Developer profile URL.

Note: Make sure you have earned all the badges as shown in above pages.



Thank you

www.eduskillsfoundation.org /EduSkillsFoundation /EduSkillsFoundation YouTube /EduSkills Foundation